

Summary

$$\vec{\nabla} \psi = \frac{1}{h_1} \frac{\partial \psi}{\partial \xi_1} \hat{\xi}_1 + \frac{1}{h_2} \frac{\partial \psi}{\partial \xi_2} \hat{\xi}_2 + \frac{1}{h_3} \frac{\partial \psi}{\partial \xi_3} \hat{\xi}_3$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$\vec{\nabla} \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\xi}_1 & h_2 \hat{\xi}_2 & h_3 \hat{\xi}_3 \\ \partial/\partial \xi_1 & \partial/\partial \xi_2 & \partial/\partial \xi_3 \\ h_1 V_1 & h_2 V_2 & h_3 V_3 \end{vmatrix}$$

CCS

SPCS

CPCS

h_1	h_2	h_3
1	1	1
1	r	$r \sin \theta$
1	ρ	1

$$\nabla^2 \psi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \xi_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \psi}{\partial \xi_1} \right) + \frac{\partial}{\partial \xi_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial \psi}{\partial \xi_2} \right) + \frac{\partial}{\partial \xi_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \psi}{\partial \xi_3} \right) \right]$$

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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

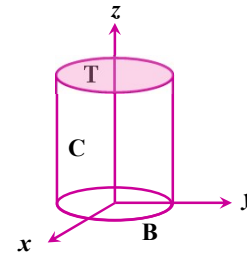
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

For **T**, $r: 0 \rightarrow 1, \phi: 0 \rightarrow 2\pi, z=1$

$$d\vec{a} = h_1 h_2 d\xi_1 d\xi_2 \hat{\xi}_3 = r dr d\phi \hat{z}$$

$$\vec{F} \cdot d\vec{a} = 10 e^{-2z} dr d\phi = 10 e^{-2} dr d\phi$$

$$\begin{aligned} \text{Flux from T} &= \int_{r=0}^1 \int_{\phi=0}^{2\pi} 10 e^{-2} r dr d\phi \\ &= 10 e^{-2} \int_{r=0}^1 r dr \int_{\phi=0}^{2\pi} d\phi = 10\pi e^{-2} \end{aligned}$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

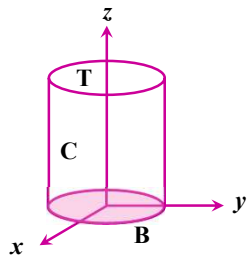
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For **B**, $r: 0 \rightarrow 1, \phi: 0 \rightarrow 2\pi, z=0$

$$d\vec{a} = -r dr d\phi \hat{z}$$

$$\vec{F} \cdot d\vec{a} = -10 e^{-2z} dr d\phi = -10 dr d\phi$$

$$\begin{aligned} \text{Flux from B} &= - \int_{r=0}^1 \int_{\phi=0}^{2\pi} 10 r dr d\phi \\ &= -10 \int_{r=0}^1 r dr \int_{\phi=0}^{2\pi} d\phi = -10\pi \end{aligned}$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

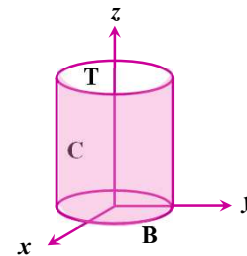
$$\text{Flux} = \oint \vec{F} \cdot d\vec{a} = \int_T \vec{F} \cdot d\vec{a} + \int_C \vec{F} \cdot d\vec{a} + \int_B \vec{F} \cdot d\vec{a}$$

For **C**, $r=1, \phi: 0 \rightarrow 2\pi, z: 0 \rightarrow 1$

$$d\vec{a} = h_2 h_3 d\xi_2 d\xi_3 \hat{\xi}_1 = r d\phi dz \hat{r}$$

$$\vec{F} \cdot d\vec{a} = 10 e^{-2z} r^2 d\phi dz = 10 e^{-2z} d\phi dz$$

$$\begin{aligned} \text{Flux from C} &= \int_{\phi=0}^{2\pi} \int_{z=0}^1 10 e^{-2z} d\phi dz \\ &= 10\pi (1 - e^{-2}) \end{aligned}$$



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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

Confirm the result using the divergence theorem.

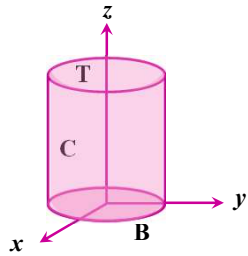
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$$\text{Flux from T} = 10 \pi e^{-2}$$

$$\text{Flux from B} = -10 \pi$$

$$\text{Flux from C} = 10 \pi (1 - e^{-2})$$

$$\text{Total Flux} = 0$$



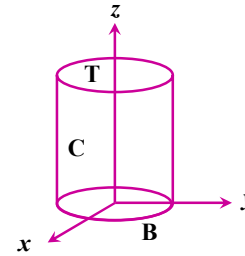
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Example

Determine the flux of a vector field $\vec{F}(r) = 10 e^{-2z} (r \hat{r} + \hat{z})$ for a cylinder $r = 1, 0 \leq z \leq 1$.

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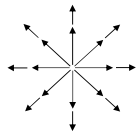
$$\begin{aligned} \vec{\nabla} \cdot \vec{F} &= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial(V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial(V_3 h_1 h_2)}{\partial \xi_3} \right] \\ &= \frac{1}{r} \left[\frac{\partial(10 e^{-2z} r r)}{\partial r} + \frac{\partial(0)}{\partial \phi} + \frac{\partial(10 e^{-2z} r)}{\partial z} \right] \\ &= \frac{1}{r} \left[10 e^{-2z} \frac{\partial r^2}{\partial r} + 0 + 10 r \frac{\partial e^{-2z}}{\partial z} \right] \\ &= \frac{1}{r} [10 e^{-2z} 2r + 0 + 10 r (-2 e^{-2z})] = 0 \quad \blacktriangleleft \end{aligned}$$



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Divergence....

Examples $\vec{V} = \frac{\hat{r}}{r^2}$ **large divergence** $\frac{h_1}{1} \frac{h_2}{r} \frac{h_3}{r \sin \theta}$



$$\begin{aligned} \vec{\nabla} \cdot \vec{V} &= \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = \vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) = \vec{\nabla} \cdot \left(\frac{x\hat{x} + y\hat{y} + z\hat{z}}{r^3} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) = 0 \quad ? \end{aligned}$$

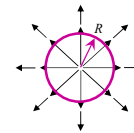
$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial(V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial(V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0$$

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Divergence....

Examples $\vec{V} = \frac{\hat{r}}{r^2}$ **Let's apply divergence theorem**



$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

$$\text{As } \vec{\nabla} \cdot \vec{V} = 0, \text{ LHS} = 0$$

$$\begin{aligned} \text{RHS : } \oint \vec{V} \cdot d\vec{a} &= \iint \left(\frac{1}{R^2} \hat{r} \right) \cdot \hat{r} R^2 \sin \theta d\theta d\phi \hat{r} \\ &= \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = 4\pi \quad ? \end{aligned}$$

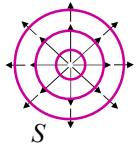
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Divergence....

Examples

$$\vec{V} = \frac{\hat{r}}{r^2}$$

Let's apply divergence theorem



$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$$

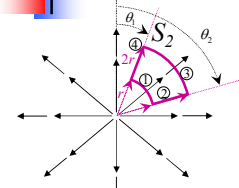
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Independent of R!

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Divergence....



$$\begin{aligned} \vec{V} &= \frac{1}{r^2} \hat{r} \\ \int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau &= \oint_{\text{surface}} \vec{V} \cdot d\vec{a} \\ \text{RHS : } \oint \vec{V} \cdot d\vec{a} &= \int_1 \vec{V} \cdot \hat{n} da + \int_2 \vec{V} \cdot \hat{n} da + \int_3 \vec{V} \cdot \hat{n} da + \int_4 \vec{V} \cdot \hat{n} da \\ &= \int_3 \frac{1}{(2r)^2} \hat{r} \cdot \hat{r} (2r)^2 \sin \theta d\theta d\phi - \int_1 \frac{1}{r^2} \hat{r} \cdot \hat{r} \sin \theta d\theta d\phi \\ &= \int_{\theta_1}^{\theta_2} \sin \theta d\theta \int_{\phi_1}^{\phi_2} d\phi - \int_{\theta_1}^{\theta_2} \sin \theta d\theta \int_{\phi_1}^{\phi_2} d\phi = 0 \\ \text{③} \quad \text{①} \quad \int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau &= \oint_{S_2} \vec{V} \cdot d\vec{a} = 0 \end{aligned}$$

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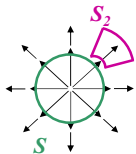
Divergence....

Examples

$$\vec{V} = \frac{\hat{r}}{r^2}$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{S_2} \vec{V} \cdot d\vec{a} = 0$$

$$\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_S \vec{V} \cdot d\vec{a} = 4\pi$$



$\vec{V}$ has a singularity at the origin

It is not differentiable at the origin

$\vec{\nabla} \cdot \vec{V}$ is not defined at the origin



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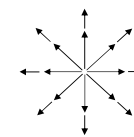
Divergence....

Examples

$$\vec{V} = \frac{\hat{r}}{r^2}$$

large divergence

$$\begin{matrix} h_1 & h_2 & h_3 \\ 1 & r & r \sin \theta \end{matrix}$$



$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial (V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial (V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0$$

$\vec{\nabla} \cdot \vec{V}$ is not defined at the origin

At all other points, $\vec{\nabla} \cdot \vec{V} = 0$

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Divergence....

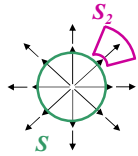
Examples

$$\vec{V} = \frac{\hat{r}}{r^2}$$

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?



$\vec{V}$ has a singularity at the origin

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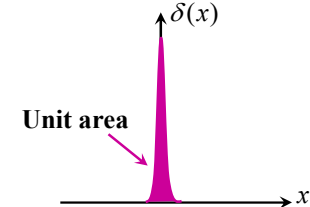
$$\vec{\nabla} \cdot \vec{V} = 4\pi \delta(\vec{r}) \quad \text{with} \quad \delta(\vec{r}) = \begin{cases} 0 & \text{if } r \neq 0 \\ \infty & \text{if } r = 0 \end{cases} \quad \text{Dirac Delta Function}$$

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One Dimensional Dirac Delta Function

$$\delta(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

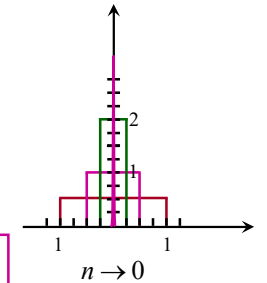


Example: 1

$$R_n(x) = \begin{cases} \frac{1}{2n} & \text{for } -n < x < n \\ 0 & \text{for } |x| > n \end{cases}$$

$$\int_{-\infty}^{\infty} R_n(x) dx = \frac{1}{2n} \int_{-n}^n dx = 1$$

$$\delta(x) = \lim_{n \rightarrow 0} R_n(x)$$

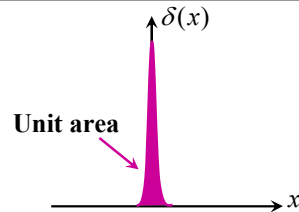


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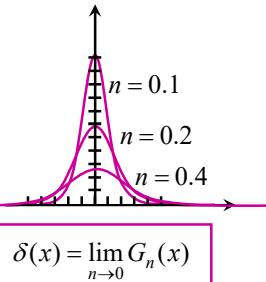


Example: 2

$$G_n(x) = \frac{1}{n\sqrt{2\pi}} \exp\left[-\frac{x^2}{2n^2}\right] \quad n > 0$$

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} \exp\left[\frac{b^2}{4a}\right]$$

$$\int_{-\infty}^{\infty} G_n(x) dx = \frac{1}{n\sqrt{2\pi}} \sqrt{\frac{\pi}{1/2n^2}} \exp\left[\frac{0}{4(1/2n^2)}\right] = 1$$



$$\delta(x) = \lim_{n \rightarrow 0} G_n(x)$$

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Properties of 1-D Dirac Delta Function

$$1. \quad \delta(x-a) = \begin{cases} 0 & \text{if } x \neq a \\ \infty & \text{if } x = a \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x-a) dx = 1$$

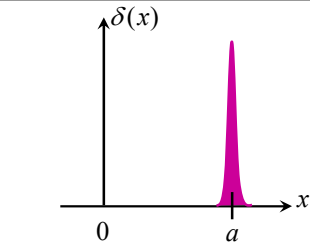
$$2. \quad f(x)\delta(x) = f(0)\delta(x)$$

$$f(x)\delta(x-a) = f(a)\delta(x-a)$$

$$3. \quad \int_{-\infty}^{\infty} f(x)\delta(x) dx = f(0) \int_{-\infty}^{\infty} \delta(x) dx = f(0)$$

$$\int_{-\infty}^{\infty} f(x)\delta(x-a) dx = f(a)$$

$$4. \quad \delta(x-a) = \delta(a-x)$$



$$5. \quad \delta[c(x-a)] = \frac{1}{|c|} \delta(x-a)$$

$$6. \quad \int_{-\infty}^{\infty} f(x)\delta'(x-a) dx = -f'(a)$$

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Three Dimensional Dirac Delta Function

$$\delta^3(\vec{r}) = \delta(x)\delta(y)\delta(z) \quad \vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\int_{\text{all space}} \delta^3(r) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x)\delta(y)\delta(z) dx dy dz = 1$$

$$\int_{\text{all space}} f(\vec{r}) \delta^3(\vec{r} - \vec{a}) d\tau = f(\vec{a})$$

Imp Note:

$$\vec{\nabla} \left(\frac{1}{r} \right) = -\frac{\hat{r}}{r^2} \quad \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = \vec{\nabla} \cdot \left(-\vec{\nabla} \left(\frac{1}{r} \right) \right) = -\nabla^2 \left(\frac{1}{r} \right) = 4\pi\delta^3(\vec{r})$$

$$\therefore \nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta^3(\vec{r})$$

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Home Work

$$J = \int_V (r^2 + 2) \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) d\tau$$

V is a sphere of radius R centered at the origin.

$$= \int_V (r^2 + 2) 4\pi\delta^3(r) d\tau$$

$$= 8\pi$$

Do it using integration by part method.

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Important Note

Q: Is a vector quantity uniquely determined by it's divergence & curl?
or

If divergence of $\vec{F}$ is given as a specified scalar function $D : \vec{\nabla} \cdot \vec{F} = D$ and curl of $\vec{F}$ as a specified vector function $\vec{C} : \vec{\nabla} \times \vec{F} = \vec{C}$, can you determine $\vec{F}$? $\vec{\nabla} \cdot \vec{C} = 0$

Lets take a simple example: Find out a vector field whose divergence and curl both are zero.

$$\vec{F}_1 = a\hat{x} + b\hat{y} + c\hat{z} \quad \vec{\nabla} \cdot \vec{F}_1 = 0 \quad \vec{\nabla} \times \vec{F}_1 = 0$$

$$\vec{F}_2 = yz\hat{x} + zx\hat{y} + xy\hat{z} \quad \vec{\nabla} \cdot \vec{F}_2 = 0 \quad \vec{\nabla} \times \vec{F}_2 = 0$$

You can find many more such possibilities...

...answer is obvious "NO"

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Important Note

Q: Is a vector quantity uniquely determined by it's divergence & curl?
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If divergence of $\vec{F}$ is given as a specified scalar function $D : \vec{\nabla} \cdot \vec{F} = D$ and curl of $\vec{F}$ as a specified vector function $\vec{C} : \vec{\nabla} \times \vec{F} = \vec{C}$, can you determine $\vec{F}$? $\vec{\nabla} \cdot \vec{C} = 0$

What would be the solution?

$$\text{Given: } \vec{\nabla} \cdot \vec{F} = D$$

$$\vec{\nabla} \times \vec{F} = \vec{C}$$

Attempt to find $\vec{F}$ is nothing but solving the two differential equations

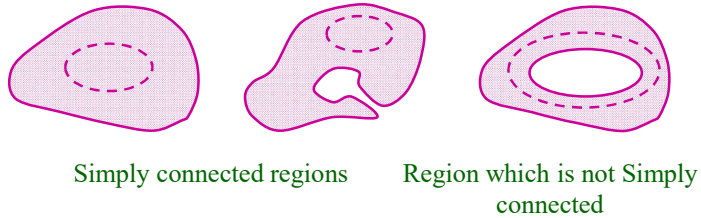
To have unique solution, we need to have appropriate boundary conditions!

In ED, the appropriate boundary condition is the requirement for the field to go to zero at infinity.

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Important Note

Consider a vector field $\vec{F}$ with its partial derivatives continuous in a simply connected region.



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Important Note

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_L \vec{F} \cdot d\vec{l}$$

Consider a vector field $\vec{F}$ with its partial derivatives continuous in a simply connected region.

Case 1: The following conditions imply each other:

- (i) $\vec{\nabla} \times \vec{F} = 0$ Every where in the region $\rightarrow \vec{F}$ is irrotational
- (ii) $\oint \vec{F} \cdot d\vec{l} = 0$ for any closed loop
- (iii) $\vec{F}$ is conservative $\rightarrow \int_a^b \vec{F} \cdot d\vec{l}$ is path independent for any given end points
- (iv) $\vec{F} = -\vec{\nabla} V$ V is termed as **scalar potential**

Note: V is not unique. Any constant can be added to V as gradient of a constant is always zero

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Important Note

$\because \vec{\nabla} \times \vec{F} = 0 \quad \therefore \oint \vec{F} \cdot d\vec{l} = 0$
 $\int_a^b \vec{F} \cdot d\vec{l} + \int_b^a \vec{F} \cdot d\vec{l} = 0$
 $\int_{a(1)}^b \vec{F} \cdot d\vec{l} - \int_{a(2)}^b \vec{F} \cdot d\vec{l} = 0$
 $\int_{a(1)}^b \vec{F} \cdot d\vec{l} = \int_{a(2)}^b \vec{F} \cdot d\vec{l}$ **Path Independent**

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_L \vec{F} \cdot d\vec{l}$$

$\int_O^P \vec{F} \cdot d\vec{l} = V(P)$
 $\vec{F} \cdot d\vec{l} = dV$
 $dV = \vec{\nabla} V \cdot d\vec{l}$
 $\vec{F} = \vec{\nabla} V$

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Important Note

Consider a vector field $\vec{F}$ with its partial derivatives continuous in a simply connected region.

Case 2: The following conditions imply each other:

- (i) $\vec{\nabla} \cdot \vec{F} = 0 \rightarrow$ Every where in the region, $\vec{F}$ is solenoidal
- (ii) $\oint \vec{F} \cdot d\vec{a} = 0$ for any closed surface
- (iii) $\int \vec{F} \cdot d\vec{a}$ is independent of the surface for any given boundary
- (iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ $\vec{A}$ is termed as **vector potential**

Note: $\vec{A}$ is not unique. Gradient of any scalar function can be added to $\vec{A}$ as curl of gradient is always zero.

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Helmholtz's Theorem

For: $\vec{\nabla} \cdot \vec{F} = D$
 $\vec{\nabla} \times \vec{F} = \vec{C}$ [$\vec{\nabla} \cdot \vec{C} = 0$]

and if D as well as $\vec{C}$ go to zero as $1/r^2 \rightarrow \infty$, and if $\vec{F}$ goes to zero as $r \rightarrow \infty$, then $\vec{F}$ is uniquely represented by

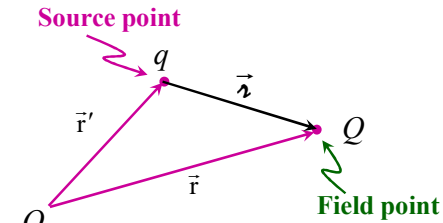
$$\vec{F} = -\vec{\nabla} V + \vec{\nabla} \times \vec{A}$$

irrotational
solenoidal

$$\vec{\nabla} \times (\vec{\nabla} V) = 0 \qquad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

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Convention, that would be followed...



$$\vec{z} = \vec{r} - \vec{r}'$$

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The Fundamental Problem of EM Theory

Q (Test Charge)

What force would be exerted on this test charge?

Complicacies are....

...how does a single charged particle interact with another single charged particle?

...does this interaction immune to the presence of other charged particles in the nbd?

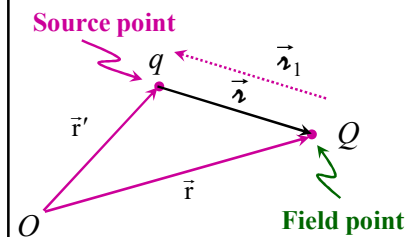
...how does the time variation of position affect this interaction?

...is it the instantaneous position of the source charge a deciding factor?

Source Charges

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Coulomb's Law



$$\vec{F}_{qQ} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{z^2} \hat{z}$$

$$\vec{z} = \vec{r} - \vec{r}'$$

$$\vec{F}_{qQ} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$\begin{aligned} \vec{F}_{Qq} &= \frac{1}{4\pi\epsilon_0} \frac{Qq}{|\vec{r}' - \vec{r}|^3} (\vec{r}' - \vec{r}) \\ &= -\frac{1}{4\pi\epsilon_0} \frac{qQ}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') \\ &= -\vec{F}_{qQ} \end{aligned}$$

ϵ_0 = permittivity of free space
 $= 8.854 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

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Coulomb's Law

A few points....

Charge is quantized, $q = Ne$, $e = 1.602 \times 10^{-19}$ C

In large objects, number of charge carriers $\sim 10^{23}$

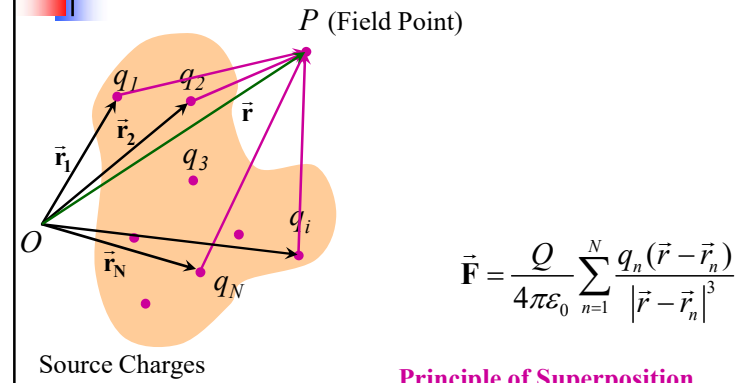
Minimum charge initially assumed to be $\pm e$, actually it is of quarks: $\pm 2e/3, \pm e/3 \dots$

...but we can not set quarks free from their bound state

Charges are conserved

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The Fundamental Problem of EM Theory



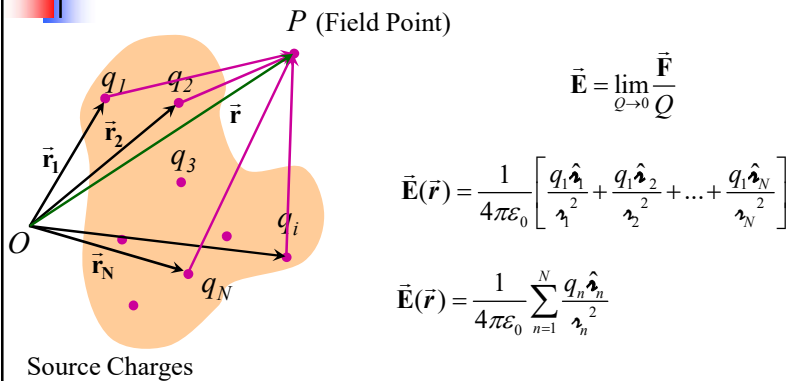
$$\vec{F} = \frac{Q}{4\pi\epsilon_0} \sum_{n=1}^N \frac{q_n(\vec{r} - \vec{r}_n)}{|\vec{r} - \vec{r}_n|^3}$$

Principle of Superposition

$$\vec{F} = \frac{Q}{4\pi\epsilon_0} \left[\frac{q_1(\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3} + \frac{q_2(\vec{r} - \vec{r}_2)}{|\vec{r} - \vec{r}_2|^3} + \dots + \frac{q_N(\vec{r} - \vec{r}_N)}{|\vec{r} - \vec{r}_N|^3} \right]$$

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Electric Field



$$\vec{E} = \lim_{Q \rightarrow 0} \frac{\vec{F}}{Q}$$

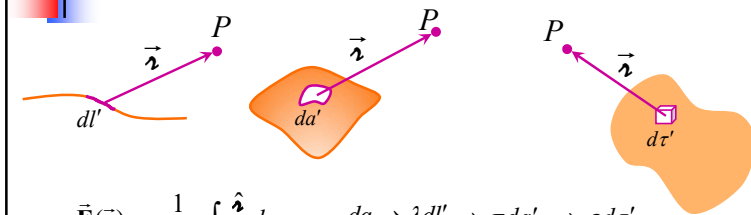
$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 \hat{r}_1}{r_1^2} + \frac{q_2 \hat{r}_2}{r_2^2} + \dots + \frac{q_N \hat{r}_N}{r_N^2} \right]$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=1}^N \frac{q_n \hat{r}_n}{r_n^2}$$

$$\vec{F} = \frac{Q}{4\pi\epsilon_0} \left[\frac{q_1(\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3} + \frac{q_2(\vec{r} - \vec{r}_2)}{|\vec{r} - \vec{r}_2|^3} + \dots + \frac{q_N(\vec{r} - \vec{r}_N)}{|\vec{r} - \vec{r}_N|^3} \right]$$

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Electric Field: Continuous Charge Distribution



$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} dq \quad dq \rightarrow \lambda dl' \rightarrow \sigma da' \rightarrow \rho d\tau'$$

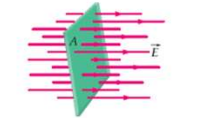
For line charge distribution $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \lambda(\vec{r}') dl'$

For surface charge distribution $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \sigma(\vec{r}') da'$

For volume charge distribution $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \rho(\vec{r}') d\tau'$

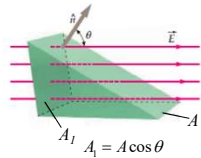
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Electric Flux

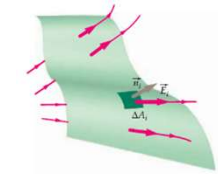


Flux through **open** surface

$$\phi = EA$$



$$\phi = EA_1 = EA_2 \cos \theta = \vec{E} \cdot \hat{n}A$$



$$d\phi = \vec{E}_i \cdot \hat{n}_i dA_i$$

$$\phi = \lim_{dA_i \rightarrow 0} \sum_i \vec{E}_i \cdot \hat{n}_i dA_i = \int_S \vec{E} \cdot \hat{n} dA$$

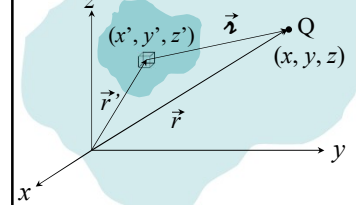
Flux through **closed** surface

$$\phi = \oint_S \vec{E} \cdot \hat{n} dA$$

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Divergence of Electric Field: Gauss's Law

General form of Coulomb's law : $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{r}}{r^2} \rho(\vec{r}') d\tau'$



$$\vec{E} = \vec{E}(x, y, z); \quad \rho = \rho(x', y', z')$$

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

$$d\tau' = dx' dy' dz'$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \frac{\hat{r}}{r^2} \rho(\vec{r}') d\tau'$$

Integration is over primed coordinates. **Let's take div w. r. t. unprimed coordinates.**

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) \rho(\vec{r}') d\tau' = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} 4\pi \delta^3(\vec{r} - \vec{r}') \rho(\vec{r}') d\tau'$$

$$= \frac{\rho(\vec{r})}{\epsilon_0} \int_{\text{all space}} \delta^3(\vec{r} - \vec{r}') d\tau' \rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho(\vec{r})}{\epsilon_0}$$

Gauss's Law

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Divergence of Electric Field: Gauss's Law

Divergence theorem : $\int_{\text{volume}} (\vec{\nabla} \cdot \vec{V}) d\tau = \oint_{\text{surface}} \vec{V} \cdot d\vec{a}$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho(\vec{r})}{\epsilon_0}$$

Electric Flux

$$\int_V \vec{\nabla} \cdot \vec{E} d\tau = \oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int_V \rho d\tau = \frac{1}{\epsilon_0} Q_{in}$$



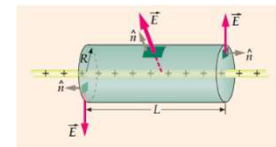
$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{in}$$

Gauss's Law

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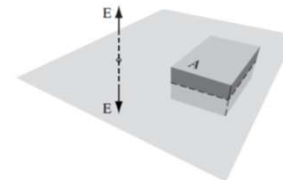
Application of Gauss's Law

Electric field due to infinite line charge



$$E_r = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{R}$$

Electric field due to infinite plane carrying uniform charge density

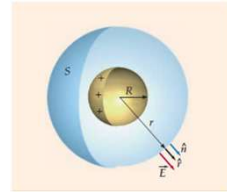


$$\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$$

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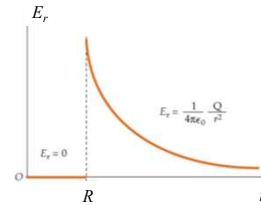
Application of Gauss's Law

Electric field due to thin spherical shell of charge

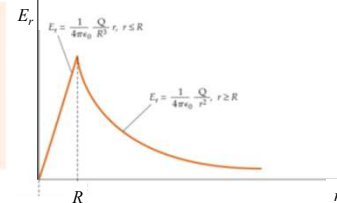
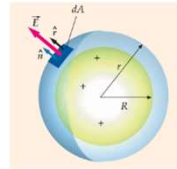
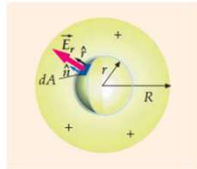


$$E_r = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad r < R$$

$$E_r = 0 \quad r > R$$



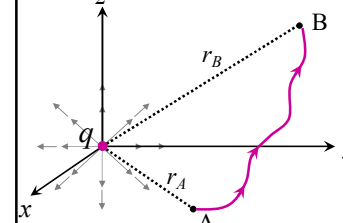
Electric field due to thin spherical shell of charge



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Curl of Electric Field

General form of Coulomb's law : $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$ $\int_A^B \vec{E} \cdot d\vec{l} = ?$



$$d\vec{l} = dr \hat{r} + r d\theta \hat{\theta} + r \sin\theta d\phi \hat{\phi}$$

$$\vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$$

$$\int_A^B \vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \int_A^B \frac{q}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_{r_A}^{r_B}$$

$$= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$

$$\int_S (\vec{\nabla} \times \vec{V}) \cdot d\vec{a} = \oint_L \vec{V} \cdot d\vec{l}$$

$$\vec{\nabla} \times \vec{E} = 0$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

$$\vec{\nabla} \times \vec{E} = \vec{\nabla} \times \vec{E}_1 + \vec{\nabla} \times \vec{E}_2 + \vec{\nabla} \times \vec{E}_3 + \dots = 0$$

$$\oint \vec{E} \cdot d\vec{l} = 0$$

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Divergence and Curl of Electric Field: Electric Potential

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E} = -\vec{\nabla} V$$

Does $\vec{E} = y \hat{x}$ represent electric field? ...answer is "NO"

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\hat{a}}{a^2} \rho(\vec{r}') d\tau'$$

$$= -\frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') \vec{\nabla} \left(\frac{1}{a} \right) d\tau' = -\vec{\nabla} \left[\frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{r}') d\tau' \right]$$

$$\rightarrow V = \frac{1}{4\pi\epsilon_0} \int_V \left(\frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) d\tau' \quad \text{Volume charge distribution}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_V \left(\frac{\lambda(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) dl' \quad \text{Line charge distribution}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_V \left(\frac{\sigma(\vec{r}')}{|\vec{r} - \vec{r}'|} \right) da' \quad \text{Surface charge distribution}$$

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Divergence and Curl of Electric Field: Electric Potential

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E} = -\vec{\nabla} V$$

$$\rightarrow \oint \vec{E} \cdot d\vec{l} = 0 \quad \text{for any closed loop}$$

$$\rightarrow \vec{E} \text{ is conservative}$$

$$\rightarrow \int_a^b \vec{E} \cdot d\vec{l} \text{ is path independent for any given end points}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{vmatrix} = 0 \rightarrow \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}; \quad \frac{\partial E_z}{\partial x} = \frac{\partial E_x}{\partial z}; \quad \frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}$$

$\vec{E}$ is a very special kind of vector...

Component of $\vec{E}$ are not really as independent as they look

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Divergence and Curl of Electric Field: Electric Potential

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \times \vec{E} = 0$$

$$\vec{E} = -\vec{\nabla} V$$

$$\oint \vec{E} \cdot d\vec{l} = 0 \quad \text{for any closed loop}$$

$\vec{E}$ is conservative

$\int_a^b \vec{E} \cdot d\vec{l}$ is path independent for any given end points

$$\int_a^b \vec{E} \cdot d\vec{l} = -\int_a^b \vec{\nabla} V \cdot d\vec{l} = -[V(b) - V(a)]$$



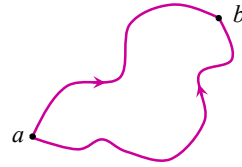
$$V(b) - V(a) = -\int_a^b \vec{E} \cdot d\vec{l}$$

$$V(b) - V(\infty) = -\int_{\infty}^b \vec{E} \cdot d\vec{l}$$

$$V(\vec{r}) = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l}$$

Superposition Principle:

$$V = V_1 + V_2 + V_3 + \dots$$

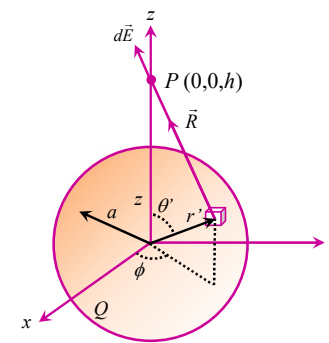


zero

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Electric Field: Continuous Charge Distribution

Home Work: Find electric field at P due to the sphere of radius a with charge Q uniformly distributed over its volume.



Answer: $\vec{E} = \frac{Q}{4\pi\epsilon_0 a^2} \hat{r}$

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